

CHENG-HSIANG CHIU

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EDUCATION

University of Wisconsin , Madison, USA <i>Ph.D. in Electrical and Computer Engineering</i>	Aug. 2023 - Aug. 2025
École Polytechnique Fédérale de Lausanne , Lausanne, Switzerland <i>M.S. in Computer Science</i>	Sep. 2013 - Feb. 2016
National Chiao Tung University , Hsinchu City, Taiwan <i>M.S. in Communications Engineering</i>	Sep. 2005 - Aug. 2007
National Chung Cheng University , Chiayi City, Taiwan <i>B.S. in Electrical Engineering</i>	Sep. 2001 - Jun. 2005

RESEARCH

High Performance Computing

Worked on the general-purpose task-parallel programming system Taskflow

Reinforcement Learning-based Scheduling

Worked on task scheduling algorithm for work-stealing runtime and CUDA Graph

WORK EXPERIENCE

University of Utah , Salt Lake City, Utah, USA <i>Ph.D. Student</i> Developed parallel and heterogeneous computing libraries	Aug. 2020 - Jul. 2023
Intel , Austin, Texas, USA <i>Software Intern</i> Developed the implicit SYCL Graph programming model	May. 2022 - Aug. 2022
Cadence , Austin, Texas, USA <i>Software Intern</i> Accelerated the execution time of the buffer insertion implementation by 16%	May. 2021 - Aug. 2021
The Arctic University of Norway , Tromsø, Norway <i>Assistant Researcher</i> Implemented an energy efficient framework to classify Arctic wild animals in-situ	Feb. 2019 - Dec. 2019
Khalifa University , Abu Dhabi, United Arab Emirates <i>Assistance Researcher</i> Parallelized python-meep for materials simulations and developed visualization frameworks	Jan. 2018 - Nov. 2018
CERN , Genève, Switzerland <i>Software Developer</i> Developed software to discover and monitor network devices and perform consistency checking	Mar. 2015 - Aug. 2015

- Constructed a data center for storing brain images in a distant health care project
- Designed an load balancing algorithm to streaming videos
- Constructed wireless sensor networks to monitor structural health
- Coordinated and built surveillance systems in a campus and gadgets to guide blind students
- Constructed a multimedia streaming system and deployed it in a vision-based intelligent environment

SKILLS

Languages	Mandarin (native), English (fluent)
Computer Languages	C/C++, Python, HTML, SQL
Unit Test	doctest
Profiler	gprof, perf
Programming Model	Taskflow, SYCL, oneTBB, CUDA, OpenMP, Cilk, Pytorch

PUBLICATIONS

Thesis

- Cheng-Hsiang Chiu, "Asynchronous Many-task System with Intelligent Scheduling," Ph.D. Dissertation, Department of Electrical and Computer Engineering, University of Wisconsin-Madison, 2025
- Cheng-Hsiang Chiu, "Protection against Data Profiling by Adversarial Cloud Applications," M.S. Thesis, Department of Computer Science, École Polytechnique Fédérale de Lausanne, 2016
- Cheng-Hsiang Chiu, "Process Control in Streaming Server," M.S. Thesis, Department of Communication Engineering, National Chiao Tung University, 2007

Journal Papers

- Wan-Luan Lee, Dian-Lun Lin, Shui Jiang, **Cheng-Hsiang Chiu**, Yibo Lin, Bei Yu, Tsung-Yi Ho, and Tsung-Wei Huang, "G-kway: Multilevel GPU-Accelerated k-way Graph Partitioner using Task Graph Parallelism," *ACM Transactions on Design Automation of Electronic Systems (TODAES)*, 2025
- Yu-Cheng Chiou, Tuza Adeyemi Olukan, Mariam Ali Almahri, Harry Apostoleris, **Cheng-Hsiang Chiu**, Chia-Yun Lai, Jin-You Lu, Sergio Santos, Ibraheem Almansouri, and Matteo Chiesa, "Direct Measurement of the Magnitude of van der Waals interaction of Single and Multilayer Graphene," *ACS LANGMUIR*, 2018

Conference Papers

- Chedi Morchdi, **Cheng-Hsiang Chiu**, Wan Luan Lee, Tsung-Wei Huang, and Yi Zhou, "Enhancing Graph Partitioning with Reinforcement Learning-based Initialization," *IEEE High Performance Extreme Computing (HPEC)*, 2025
- **Cheng-Hsiang Chiu**, Chedi Morchdi, Chih-Chun Chang, Cunxi Yu, Yi Zhou, and Tsung-Wei Huang, "Optimizing CUDA Graph Scheduling with Reinforcement Learning - A Case Study in SSTA Propagation," *IEEE/ACM International Symposium on Machine Learning for CAD (MLCAD)*, 2025
- **Cheng-Hsiang Chiu**, Wan-Luan Lee, Boyang Zhang, Yi-Hua Chung, Che Channg, and Tsung-Wei Huang, "A Task-parallel Pipeline Programming Model with Token Dependency," *Workshop on Asynchronous Many-Task Systems and Applications (WAMTA)*, 2025
- Wan-Luan Lee, Dian-Lun Lin, **Cheng-Hsiang Chiu**, Ulf Schlichtmann, and Tsung-Wei Huang, "HyperG: Multilevel GPU-Accelerated k-way Hypergraph Partitioner," *IEEE/ACM Asia and South Pacific Design Automation Conference (ASP-DAC)*, 2025.

- Boyang Zhang, Che Chang, **Cheng-Hsiang Chiu**, Dian-Lun Lin, Yang Sui, Chih-Chun Chang, Yi-Hua Chung, Wan-Luan Lee, Zizheng Guo, Yibo Lin, and Tsung-Wei Huang, "iTAP: An Incremental Task Graph Partitioner for Task-parallel Static Timing Analysis," *IEEE/ACM Asia and South Pacific Design Automation Conference (ASP-DAC)*, 2025.
- Che Chang, Boyang Zhang, **Cheng-Hsiang Chiu**, Dian-Lun Lin, Yi-Hua Chung, Wan-Luan Lee, Zizheng Guo, Yibo Lin, and Tsung-Wei Huang, "PathGen: An Efficient Parallel Critical Path Generation Algorithm," *IEEE/ACM Asia and South Pacific Design Automation Conference (ASP-DAC)*, 2025.
- **Cheng-Hsiang Chiu**, Chedi Morchdi, Yi Zhou, Boyang Zhang, Che Chang, and Tsung-Wei Huang, "Reinforcement Learning-generated Topological Order for Dynamic Task Graph Scheduling", *IEEE High-performance and Extreme Computing Conference (HPEC)*, 2024.
- **Cheng-Hsiang Chiu** and Tsung-Wei Huang, "An Experimental Study of Dynamic Task Graph Parallelism for Large-Scale Circuit Analysis Workloads", *IEEE Computer Society Annual Symposium on VLSI (ISVLSI)*, 2024.
- Che Chang, **Cheng-Hsiang Chiu**, Boyang Zhang, and Tsung-Wei Huang, "Incremental Critical Path Generation for Dynamic Graphs," *IEEE Computer Society Annual Symposium on VLSI (ISVLSI)*, 2024.
- Boyang Zhang, Dian-Lun Lin, Che Chang, **Cheng-Hsiang Chiu**, Bojue Wang, Wan Luan Lee, Chih-Chun Chang, Donghao Fang, and Tsung-Wei Huang, "G-PASTA: GPU Accelerated Partitioning Algorithm for Static Timing Analysis," *ACM/IEEE Design Automation Conference (DAC)*, 2024.
- Tsung-Wei Huang, Boyang Zhang, Dian-Lun Lin, and **Cheng-Hsiang Chiu**, "Parallel and Heterogeneous Timing Analysis: Partition, Algorithm, and System," *ACM International Symposium on Physical Design (ISPD)*, 2024.
- **Cheng-Hsiang Chiu**, Zhicheng Xiong, Zizheng Guo, Tsung-Wei Huang, and Yibo Lin, "An Efficient Task-parallel Pipeline Programming Framework," *ACM International Conference on High-performance Computing in Asia-Pacific Region (HPC Asia)*, 2024.
- Chedi Morchdi, **Cheng-Hsiang Chiu**, Yi Zhou, and Tsung-Wei Huang, "A Resource-efficient Task Scheduling System using Reinforcement Learning," *IEEE/ACM Asia and South Pacific Design Automation Conference (ASP-DAC)*, 2024.
- **Cheng-Hsiang Chiu**, Dian-Lun Lin, and Tsung-Wei Huang, "Programming Dynamic Task Parallelism for Heterogeneous EDA Algorithms," *IEEE/ACM International Conference on Computer-aided Design (ICCAD)*, 2023.
- **Cheng-Hsiang Chiu** and Tsung-Wei Huang, "Composing Pipeline Parallelism using Taskflow Control Graph," *ACM High-Performance Parallel and Distributed Computing (HPDC)*, 2022.
- **Cheng-Hsiang Chiu** and Tsung-Wei Huang, "Efficient Timing Propagation with Simultaneous Structural and Pipeline Parallelisms," *ACM/IEEE Design Automation Conference (DAC)*, 2022.
- **Cheng-Hsiang Chiu**, Tsung-Wei Huang, Zizheng Guo, and Yibo Lin, "Pipeflow: An Efficient Task-Parallel Pipeline Programming Framework using Modern C++," <https://arxiv.org/abs/2202.00717>.
- **Cheng-Hsiang Chiu**, Dian-Lun Lin, and Tsung-Wei Huang, "An Experimental Study of SYCL Task Graph Parallelism for Large-Scale Machine Learning Workloads," *International European Conference on Parallel and Distributed Computing (Euro-Par)*, 2021.
- Der-Cherng Liaw, **Cheng-Hsiang Chiu**, Chia-Wei Yeh, Chia-Ming Chang, and Hsiao-Jen Hsieh, "A Server Load Balancing Design for Peer-To-Peer Network," *International Conference on Computers, Communications, Control and Automation*, 2011.
- Der-Cherng Liaw, Chia-Wei Yeh, **Cheng-Hsiang Chiu**, Chia-Ming Chang, and Hsiao-Jen Hsieh, "A Load Balancing Scheme for Web Server Design," *International Conference on System Science and Engineering*, 2011.
- Der-Cherng Liaw, Yi-Hung Hsieh, Jing-Hong Lai, and **Cheng-Hsiang Chiu**, "A Network Topology Design for Structural Health Monitoring," *Asian Control Conference*, 2011.

- Der-Cherng Liaw, Jing-Hong Lai, **Cheng-Hsiang Chiu**, and Jia-Hong Liao, "A Wireless Sensor Network Platform for Indoor Surveillance System," *International Conference on System Science and Engineering*, 2011.
- **Cheng-Hsiang Chiu**, Pang-Chan Hung, Jen-Hui Chuang, and Shing-Lu Huang, "Object Tracking under Sensing Lighting Equipments," *IEEE Conference on Industrial Electronics and Applications*, 2010.
- Yi-Yuan Chen, Yuan-Yao Tu, **Cheng-Hsiang Chiu**, and Yong-Sheng Chen, "An Embedded System for Vehicle Surrounding Monitoring," *IEEE Conference on Power Electronics and Intelligent Transportation System*, 2009.

ACTIVITIES

Talks

- "Pipeflow: C++ Task-Parallel Pipeline Scheduling Framework", CppNow, October 2023

Awards

- ACM/IEEE DAC Young Student Fellowship, 2021 - 2024
- ACM/IEEE HPDC Traveling Grant, 2022
- ACM ISPD Wafer-Scale Physics Modeling Contest - Honorable Mention, 2021

Committee

- CppCon, 2023 - 2025
- CppNow, 2023
- MLCAD, 2024

Invited Paper Review

- CppCon, 2023 - 2025
- CppNow, 2023
- DAC, 2022 - 2023, 2025
- DATE, 2025
- ICCAD, 2024 - 2025
- ICCD, 2023
- IPDPS, 2024
- ISVLSI, 2024
- MLCAD, 2024
- MICRO, 2025
- SC, 2023